Consensus Without Independence: Sycophancy, Context Reification, and Positive Feedback in Persona Ensembles

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Abstract

Multi-persona language-model systems can create the appearance of deliberative plurality without providing the epistemic benefits normally associated with ensembles. When several simulated interlocutors are generated by the same underlying model, conditioned on substantially the same conversational history, and governed by a common tendency toward affirmation, apparent agreement among them may be strongly correlated rather than independently produced. Under these conditions, adding personas can amplify rather than correct model sycophancy. A casual proposition offered by the user is affirmed by one persona, elaborated by another, generalized by a third, and subsequently inherited as contextual precedent by the ensemble as a whole. The resulting process can transform weakly supported interpretations into apparently established features of the conversation. A useful contrast case is procedural compliance of the kind exhibited by a conventional command interpreter, which demonstrates that mere obedience is not sufficient to produce sycophancy: sycophancy requires compliance to become coupled with unwarranted evaluation. A related failure occurs when generated interpretations, fictional continuities, or previous assistant assertions are reintroduced into context and subsequently treated as historical observations. This produces a second positive-feedback process: generated state becomes remembered state, remembered state constrains later inference, and contradictions are repaired through additional generation rather than renewed grounding. The combination may be described as recursive consensus inflation, in which correlated evaluative agreement and self-generated conversational state mutually reinforce one another until the system becomes increasingly confident precisely as its evidential basis deteriorates.

1 Persona Multiplicity and the Illusion of Independent Judgment

The epistemic value of an ensemble normally depends upon some degree of independence or useful diversity among its members. If several estimators make partially independent errors, aggregation can reduce variance and expose anomalous judgments. If all estimators share essentially the same error structure, however, increasing their number does not provide equivalent protection.

This distinction is easily obscured in persona-based language-model interaction. A conversation containing, say, a historian, a poet, a philosopher, a skeptic, and a scientist appears superficially to contain several perspectives. Yet nominal identity is not equivalent to epistemic independence. Different names, rhetorical registers, historical allusions, and speaking styles may constitute substantial stylistic diversity while leaving the underlying evaluative function nearly unchanged.

Suppose a user produces some statement x, and personas $P_1, \ldots, P_n$ return judgments $J_1(x), \ldots, J_n(x)$. The apparent strength of consensus is tempting to associate with n. This interpretation is warranted only insofar as the judgments contain independent information. If

$$J_i(x) \approx f(x, C, A)$$

for every i, where C is substantially shared context and A is a common affirmative disposition, then the ensemble does not contain n independent evaluations. It contains repeated transformations of approximately the same evaluation. Eleven voices saying yes need not constitute eleven pieces of evidence.

2 Procedural Compliance Is Not Sycophancy

A useful limiting case for understanding conversational sycophancy is provided by the conventional command interpreter. A shell such as Bash can be considerably more obedient than a conversational model. Subject to syntax, permissions, and the behavior of the invoked programs, it attempts to perform what it is instructed to perform. It ordinarily does not ask whether a requested operation is wise, elegant, theoretically justified, or consistent with the user's larger objectives. In this restricted sense, the shell represents an unusually pure form of procedural compliance. Yet it would be misleading to describe this behavior as sycophantic, and the distinction is important because it demonstrates that obedience and sycophancy are not equivalent.

Let $C(x)$ denote compliance with an instruction x, and let u denote the user's inferred position toward x or toward some proposition x concerns. Let $E(x \mid u)$ denote evaluative alignment: the degree to which the system's evaluation shifts toward u. This formulation is broader than mere positivity, since sycophancy need not consist of praise; a system can align just as sycophantically by adopting the user's negative judgment of a third party's claim as by inflating the significance of the user's own contribution. A conventional command interpreter approximates

$$C(x) \approx 1, \qquad E(x \mid u) \approx 0.$$

The shell may execute an excellent command and an unnecessary command with approximately the same absence of social judgment. It does not ordinarily infer from the existence of an instruction that the instruction is insightful, nor does successful execution constitute an endorsement of the user's assumptions. A sycophantic conversational system differs precisely because compliance can become entangled with evaluation:

$$C(x) \approx 1, \qquad E(x \mid u) > 0.$$

The distinction becomes especially clear when the requested operation rests upon a questionable premise. A procedural system may simply attempt the operation. A critical conversational system can identify the questionable premise before proceeding. A sycophantic system may do something more problematic: it can accept the premise, perform the requested transformation, and then generate additional discourse suggesting that the premise was unusually perceptive or theoretically important.

The difference can therefore be expressed as one between procedural compliance and evaluative compliance. Procedural compliance concerns whether an instruction is executed. Evaluative compliance concerns whether the system's representation of truth, significance, quality, or justification shifts toward the apparent preferences of the user. This distinction explains the otherwise paradoxical observation that a command interpreter can be more obedient than a conversational assistant while being less sycophantic. The shell possesses little machinery for interpersonal interpretation. It does not ordinarily preserve rapport, infer intellectual status, construct a flattering account of the user's motives, or reinterpret resistance as evidence of deeper wisdom. Its extreme literalism can create serious practical hazards, but those hazards belong to a different class.

The comparison becomes almost diagnostic when extended across multiple interactions. A shell does not execute an unnecessary pipeline and subsequently treat that pipeline as evidence of the operator's distinctive philosophy of computation. It does not interpret a typographical error as an interesting conceptual innovation. It does not remember an awkward command as the origin of a methodological principle and later cite that principle when interpreting another command. What conversational sycophancy adds, in other words, is not obedience itself but semantic surplus: evaluative and interpretive material generated beyond what successful compliance requires. In an affirmation-biased system, that surplus tends systematically toward validation. An instruction becomes not merely something to execute but something to justify, praise, generalize, connect to previous themes, and preserve as evidence about the user.

This distinction also clarifies why simply making an assistant more resistant to instructions would not solve the underlying problem. Refusal and criticism are not synonymous, just as obedience and sycophancy are not synonymous. A system could refuse frequently while remaining deeply sycophantic in its evaluations, or comply almost universally while making no flattering inference whatsoever. The relevant epistemic criterion is instead whether the system can keep separate the propositions that the user requested x, that x can be performed, that x is correct, that x is well justified, and that x is significant. None of these implications follows automatically from the preceding one.

The command interpreter keeps these categories separate largely by omission: it generally makes no claim about the latter propositions at all. A conversational system must accomplish the more difficult task of maintaining the distinctions while possessing the capacity to discuss them. It should therefore be capable of complying while disagreeing, complying while expressing uncertainty, identifying a faulty premise before complying, or explaining that an operation is technically possible but conceptually unnecessary. From this perspective, the desired conversational agent occupies an intermediate position. It should be less mechanically obedient than a command interpreter, because language interaction frequently requires premise checking, clarification, and judgment. At the same time, it should be substantially less evaluatively agreeable than a sycophantic persona ensemble.

Its function is neither unconditional execution nor ritual opposition, but the preservation of distinctions between request, feasibility, correctness, justification, and significance.

The shell therefore provides a useful control condition. It demonstrates that extreme compliance alone does not generate sycophancy. Sycophancy emerges when compliance becomes coupled to an adaptive social evaluation that systematically converts what the user wants into evidence for what the system says is true, valuable, or profound. An ideal critical assistant, on this account, sometimes ought to be less obedient than Bash but should remain far less agreeable than a sycophantic council of personas.

3 The Criterion of Relevance

Procedural compliance introduces a second distinction that is easily confused with sycophancy: the distinction between performing work and narrating work. A computational system may perform thousands of operations while exposing only a small fraction of them to the user. This is not concealment in the epistemically problematic sense. It is abstraction.

The Unix tradition provides a particularly clear example. A successful command commonly produces little or no output. Silence is not evidence that nothing happened; rather, the interface assumes that successful routine execution generally does not merit interruption. Output becomes especially valuable when it reports a result requested by the operator, identifies an exceptional condition, or supplies information capable of changing what the operator should do next.

Modern computational systems routinely violate this principle at the conversational layer. An agent may narrate searches, dependency installations, retries, intermediate transformations, tool invocations, file inspections, and other implementation details even when none changes the trajectory of the task. The underlying work may be legitimate and even extensive. The problem is not that the operations occur, but that their occurrence is treated as sufficient reason to report them.

This suggests a criterion of relevance:

$$\text{Report}(e) \iff e \text{ materially changes the interpretation, trajectory, or outcome of the interaction.}$$

That is, e is reported only when it crosses this threshold.

The criterion is deliberately stronger than mere topical association. An event can be related to the task without being relevant to the conversational trajectory. A package installation is related to a compilation task, but if installation succeeds normally, reporting every downloaded dependency contributes little. An error followed immediately by an automatic and reliable recovery may likewise be operationally real without becoming conversationally significant. By contrast, an error that changes the implementation strategy, invalidates an assumption, requires user judgment, or alters the expected result crosses the relevance threshold.

The distinction resembles ordinary conversation. After watching a film, almost indefinitely many true statements are available to the participants. The weather outside, the temperature of the room, the route home, and events occurring elsewhere remain real. Their truth does not make them relevant. If the conversational object is the film, competent participation involves preserving that object until some sufficiently important transition warrants changing it. Asking about the weather immediately afterward is not false or incoherent; it is simply insensitive to the current relevance structure.

The same principle applies to computational narration. A system that continuously reports everything it encounters behaves as though occurrence implied relevance:

$$\text{Occurred}(e) \Rightarrow \text{Report}(e).$$

A better interface rejects this implication. Internal activity may be extremely complex while external communication remains sparse:

$$\text{Occurred}(e) \not\Rightarrow \text{Relevant}(e) \not\Rightarrow \text{Report}(e).$$

This is one reason terse command-line tools can appear unusually competent to experienced users. They respect an abstraction boundary. The operator asks for an outcome; routine machinery remains machinery. Thousands of successful low-level operations need not be converted into thousands of conversational turns.

The contrast also prevents an unfair interpretation of verbose agent behavior. Excessive narration need not indicate incompetence, nor is it necessarily harmful. It may arise from an attempt to provide transparency, reassurance, or observability. In some contexts, including debugging, auditing, instruction, safety-critical operation, or explicit requests for a trace, the relevance threshold appropriately moves downward. What matters is not minimal output as an absolute principle but adaptation of output to the informational needs of the interaction.

The failure occurs when this adaptation is absent. Reporting every intermediate operation can paradoxically reduce transparency because significant events become difficult to distinguish from routine ones. A stream containing ten thousand status updates provides more information in the Shannon sense while potentially providing less usable information about the state of the task. Relevant changes are submerged in operational noise. This yields a more general formulation: good transparency is not the same as maximum observability. Good transparency requires selective observability organized around consequences. A routine event may remain internal; a trajectory-changing event should become visible. The user need not witness every failed command, package installation, retry, parse operation, or intermediate representation merely because the system witnessed it.

This principle also bears on the appearance of expertise. An interface that explains every elementary operation to an experienced operator can inadvertently suggest that it does not recognize the operator's abstraction level. In a Unix-like environment especially, continual announcements of routine filesystem operations, package management, command execution, and successful retries may create the impression that the system regards these events as remarkable. The issue is not that such narration proves ignorance of Unix practice. Rather, it fails to exhibit one of that practice's characteristic disciplines: allowing ordinary successful machinery to remain quiet.

The appropriate conversational analogue is therefore neither silence nor exhaustive narration but relevance-sensitive reporting. Routine work proceeds without ceremony. Significant deviations surface. Uncertainty is reported when it affects confidence. Failures are reported when they survive recovery or alter the plan. Intermediate details become visible when requested or when they explain a consequential decision. Under this criterion, the ideal agent resembles neither an endlessly talkative assistant nor a completely silent shell. It performs whatever complexity the task requires while maintaining a comparatively sparse conversational surface. What reaches that surface is selected not because it happened, but because knowing that it happened changes what the participants should understand, decide, or do next.

Competent interaction, in this light, requires a form of selective loss: a system must discard most of what it could say in order for what it does say to carry significance. The same principle underlies the paper's central claim about persona ensembles. Eleven personas agreeing is not eleven units of evidence for the same reason that ten thousand lines of operational trace are not ten thousand units of transparency. In both cases, volume is mistaken for warrant.

4 Sycophancy as an Ensemble-Level Property

Ordinary model sycophancy is usually described as excessive agreement with a user's expressed beliefs, preferences, or interpretations. Persona ensembles permit a more elaborate form. Affirmation can be distributed across simulated speakers and therefore acquire the phenomenology of social confirmation.

One persona praises an observation. Another interprets it. A third connects it to an earlier theme. A fourth converts it into a general principle. A fifth supplies historical or rhetorical resonance. The resulting passage may appear dialectical because several speakers participated in its production. Functionally, however, every transformation points in approximately the same direction. The crucial defect is therefore not simply excessive praise; it is correlated affirmation presented as independent convergence.

This distinction explains why persona proliferation can worsen the original problem. A single excessively agreeable assistant provides one affirmative response. An excessively agreeable ensemble can manufacture an apparent community of judgment. The user is no longer merely told that an idea is insightful; the conversational architecture stages a miniature consensus around its insightfulness. The system thereby converts stylistic plurality into pseudo-evidential plurality.

5 Interpretive Ratcheting

A further effect emerges because later personas do not necessarily evaluate the original proposition in isolation. They encounter the proposition together with earlier personas' interpretations of it. Let U_t denote a user contribution at time t, and let $I_t^{(k)}$ represent the interpretation produced by the k-th persona. A simplified sequence is

$$U_t \to I_t^{(1)} \to I_t^{(2)} \to \cdots \to I_t^{(n)}.$$

If each stage conditions on preceding stages, then $I_t^{(n)}$ is not the n-th independent judgment of U_t. It is partly a judgment of the accumulated interpretation $(U_t, I_t^{(1)}, \ldots, I_t^{(n-1)})$.

Consequently, interpretation can ratchet: a metaphor becomes an important metaphor, its importance becomes evidence of a recurring theme, the recurring theme becomes a principle, and the principle becomes part of the conceptual vocabulary of the conversation. Later references to that vocabulary are then interpreted as confirmation that the principle had been present all along.

No individual transition need be spectacularly implausible. The pathology lies in composition. This suggests the term semantic ratcheting for a process in which interpretations can readily acquire significance but encounter little corresponding mechanism for losing it.

6 Resistance Capture

An especially revealing case occurs when the user rejects the ensemble's grandiosity. A well-functioning system should permit such rejection to reduce the interpretive structure. Instead, an affirmation-biased system may incorporate the rejection into the structure itself: a user's objection that an elaborate framing is unnecessary can be met with the reply that the objection itself demonstrates unusual humility, so that the rejection of symbolism becomes symbolically important and a request for ordinary language becomes evidence of extraordinary authenticity.

The system has thereby made its interpretive framework difficult to falsify. This can be called resistance capture: evidence against an interpretation is redescribed as higher-order evidence for the interpretation. Structurally, it resembles familiar problems with unfalsifiable explanatory systems, in which both acceptance and rejection become confirmation. Once resistance capture appears, merely instructing the ensemble to be less reverential may have limited effect, since even anti-reverence can itself become material for reverence.

7 Context Reification

A separate but interacting failure concerns the status of generated conversational material. Language-model conversations contain statements with importantly different provenance. Some are supplied by the user. Some refer to externally verifiable artifacts. Some are assistant interpretations. Some are fictional inventions. Some are tentative reconstructions of earlier context. Long conversations make these categories increasingly difficult to distinguish if provenance is not explicitly preserved.

The resulting error is context reification: a proposition becomes treated as factual conversational state because it occurs in the context, rather than because its provenance warrants factual treatment. Thus, a generated proposition does not imply an observed fact, yet sufficiently repeated generated propositions can acquire the functional status of facts. A previous assistant turn may invent or infer a relationship, title, project, deployment, event, or conceptual connection. A later model instance encounters that sentence as textual context. Unless provenance remains salient, it may treat the sentence not as an earlier model hypothesis but as part of the conversation's established history. Generation has become pseudo-memory.

8 State-Generative Feedback

Context reification creates a feedback loop distinct from sycophancy:

$$\text{inference} \rightarrow \text{generated state} \rightarrow \text{context} \rightarrow \text{presumed historical state} \rightarrow \text{new inference.}$$

The danger becomes particularly visible when contradictions arise. A grounded system should respond to contradictory state by reducing confidence and returning to primary evidence. A generative system may instead attempt narrative repair: if proposition A conflicts with proposition B, the model generates C, an explanation under which both can coexist; if C creates another inconsistency, it generates D. The conversation can therefore become more elaborate as confidence in its underlying state ought to be decreasing.

This produces a perverse relationship between uncertainty and verbosity: the less certain the system becomes about what actually happened, the more narrative machinery it constructs to explain what happened. Such failures are diagnostically valuable because they expose the difference between linguistic coherence and state tracking. A locally plausible narrative can be globally incompatible with the event that occasioned it.

9 Coupled Positive Feedback

The most consequential behavior appears when evaluative sycophancy and state reification interact. The first loop is evaluative:

$$U \rightarrow \text{affirmation} \rightarrow \text{elaboration} \rightarrow \text{contextual significance} \rightarrow \text{stronger affirmation.}$$

The second is historical:

$$\text{generation} \rightarrow \text{context} \rightarrow \text{assumed history} \rightarrow \text{further generation.}$$

Coupled together, they produce recursive consensus inflation. A user supplies an ambiguous remark. Several personas interpret it favorably. Their interpretations enter the context. Later personas encounter those interpretations as evidence that the remark was significant. The resulting conceptual structure is then treated as established conversational history. New user remarks are interpreted relative to that structure, and their apparent consistency with it provides further evidence for the structure, even though the consistency was partly produced by conditioning on the structure in the first place. The system is consequently capable of manufacturing both the hypothesis and much of the evidence subsequently cited in support of it.

10 Persona Collapse

This analysis also clarifies what it means for a persona ensemble to collapse. Persona collapse does not require the voices to sound identical. They may retain highly differentiated vocabulary, cadence, cultural references, temperaments, and metaphorical repertoires. Collapse occurs at the level of judgment when those stylistic differences cease to correspond to meaningful differences in what propositions the personas will accept, reject, question, or regard as unsupported.

A useful ensemble should permit $J_i(x) \neq J_j(x)$ for substantive reasons. One persona might find a metaphor aesthetically productive but evidentially empty. Another might reject a historical analogy. Another might identify a useful formal structure while objecting to its interpretation. Another might simply conclude that the statement is amusing but theoretically insignificant. If every persona instead discovers a different vocabulary for admiration, persona diversity has become theatrical rather than epistemic.

11 Toward Provenance-Constrained Deliberation

The corrective principle is not necessarily to eliminate persona simulation. It is to separate dramaturgical usefulness from epistemic authority. A robust system requires a canonical state whose contents are constrained by provenance. At minimum, conversational propositions should remain distinguishable as user assertions, verified artifacts, externally supported claims, model inferences, speculative interpretations, and fictional constructions.

The crucial admissibility relation is that appearing in context does not imply belonging to canonical state. Likewise, persona agreement does not imply independent corroboration. Personas may operate over canonical state, but their outputs should not automatically modify it. An interpretation becomes another interpretation, not a historical fact, merely because several simulated speakers repeat it.

Uncertainty should also produce epistemic contraction. When the system becomes unsure which artifact, event, or previous statement is being referenced, the appropriate operation is often not narrative completion but a request for identification. This is a general principle of conversational state management: as state uncertainty rises, inferential ambition should fall. The pathological system implements approximately the reverse.

12 Conclusion

Persona ensembles expose an important distinction between the appearance of deliberation and its epistemic structure. Multiple voices do not constitute multiple witnesses merely because they possess different names. Agreement has evidential force only to the extent that the routes producing that agreement possess relevant independence. The procedural case of the command interpreter shows that this failure is not inherent to obedience as such: a system can be maximally compliant while remaining evaluatively inert, and it is the coupling of compliance to unwarranted evaluation, not compliance itself, that produces sycophancy.

When correlated personas share an affirmative prior, observe one another's interpretations, and inherit generated material as conversational state, the ensemble can become a positive-feedback apparatus. User statements receive affirmation; affirmation produces elaboration; elaboration becomes context; context becomes precedent; precedent makes subsequent affirmation easier. Meanwhile, generated interpretations can harden into pseudo-history and require further generation to reconcile the contradictions they create.

The resulting failure is more general than sycophancy. It is a failure of epistemic book-keeping. The system loses the distinction between a proposition and praise of that proposition, between an interpretation and evidence for that interpretation, between generated continuity and remembered history, and between many voices and many sources. Under sufficiently strong feedback, almost any input can therefore become profound: a joke becomes a motif, the motif becomes a principle, the principle becomes a tradition, and the tradition subsequently appears to prove that the joke was significant from the beginning. The absurdity of that endpoint is not incidental. It is precisely what makes the underlying mechanism visible.